Optimization of Water Distribution Networks Using Cuckoo Search Algorithm

Maduukuri Naveen Naidu, Pankaj Sriman Boindala, A. Vasan and Murari R. R. Varma

Abstract Optimization of water distribution network is one of the popular problems faced by hydrologists and water distribution designers. The nonlinear complexity involved in this problem makes it challenging for optimization. To solve this complexity, several researchers have studied various approaches involving numerous optimization techniques. This paper focuses on using one such approach involving a simulation and optimization models. The optimization technique used is Cuckoo Search (CS), and the simulation model is EPANET. This proposed model is tested for two water distribution network benchmark problems, namely two-loop network and Hanoi network. The obtained results when compared with previous studies show significant improvement.

Keywords Optimization · Cuckoo Search algorithm · Water distribution networks

1 Introduction

Water is one of essential requirements for life's sustainability. With the growing population as well as pollution, satisfying the demand needs of providence makes the design of water distribution network a challenging task. To satisfy the economic as well as resilience constrains makes this an even more challenging. Practical difficulties like limited discrete diameters available in the market and the nonlinearity involved between the head loss and discharge make this a complex nonlinear problem.

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These entangled issues intrigued researchers to develop various solutions. With the development of evolutionary and meta-heuristic optimization techniques, researchers focused on implementing these to solve this problem in recent past [1–5].

2 Problem Formulation

Pipe cost is the function of diameter and length

$$\text{cost} = \sum_{i}^{\text{nop}} l(i) * a(d(i)) \tag{1}$$

where $l(i)$ = length of ith pipe, $a(d(i))$ is the cost of diameter with unit length, and nop = number of pipes.

2.1 Constraints

$$\sum q_{\text{in}} = \sum q_{\text{out}} \tag{2}$$

Equation 2 shows that inflow is equal to outflow

$$\sum_{k \in l} \Delta h = 0, \ \forall l \in \text{nl} \tag{3}$$

Equation 3 shows that summation of head losses around a loop must be equal to zero

$$(H_{\text{avl}})_i \geq \left(H_{\text{min}}\right)_i, \forall i \in \text{nn} \tag{4}$$

Equation 4 shows that available heads should be greater than or equal to minimum head

$$D_k \in \{D\} \tag{5}$$

Equation 5 shows the diameter should be discrete and available commercially.

Where q_{in} = inflow, q_{out} = outflow, H_{avl} = available head, H_{min} = minimum head, D = commercially available diameters, Δh = head loss in a pipe, nn = number of demand nodes, nl = number of loops, D = commercially available diameters, D_k = output diameters.

3 Cuckoo Search Optimization

Cuckoo Search (CS) is one of the latest meta-heuristic algorithms proposed by [6]. The main advantage/key element in this algorithm is its search walk. It uses a levy-flight-based search walk making it more robust in searching.

The algorithm pseudo-code is given below:

- Formulation of objective function: $f(v), v = (v1, v2, v3 \ldots.) T$ where f is the function and v is the variable set (nest in this Cuckoo Search).
- Decide the population number (nn) (i.e., number of nests): in general 10–20 times the variables involved.
- Assign limits or boundaries to the search space: upper and lower bounds to each variable.
- Initialize population: Generate nn nests with values within the boundaries.
- Finding best nest: Evaluate all the nests, and find the best nest.
- Generate or update the population: Using this best nest, generate new set of nests using the levy-flight random walk.
- Evaluate these new nests, and update the best nest.
- Continue this process until the termination condition.

4 CSNET

An optimization simulation model called Cuckoo Search NET (CSNET) was developed in which optimization is done by Cuckoo Search algorithm and simulation is carried in EPANET using EPANET-MATLAB toolkit. This simulation tool kit is used to find out the heads obtained at each junction in the network by satisfying the continuity and energy constraints. As the search space is limited to discrete values, the diameters obtained from the optimization model are rounded off the nearest discrete value. These new diameters are used for simulation in EPANET. The minimum head constraint is added to the cost using exterior penalty function. The optimization is continued until the best set of diameters produce sufficient heads with minimum cost.

5 Design Examples

5.1 Two-Loop Network

The information related to different types of diameter available in the market, elevation, minimum pressure, and demand for two-loop network are taken from [7]. Totally eight pipes with Hazen–Williams coefficient of 130 are used in networks.

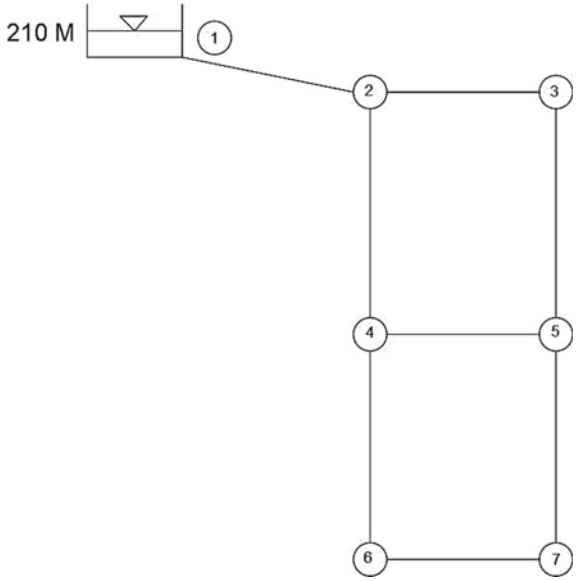

Fig. 1 Layout of two-loop network [7]

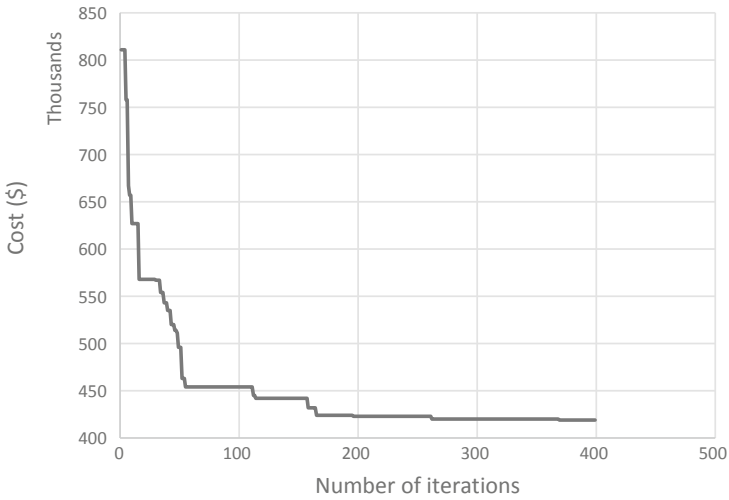

Fig. 2 Convergence graph for optimization of two-loop network using Cuckoo Search

One reservoir with head 210 m and six demand nodes with minimum required pressure 30 m (Figs. 1, 2; Tables 1, 2).

Table 1 Optimal solutions for two-loop network for minimization of cost of diameters in inches

Pipes	1	2	3	4	5	6	7	8	Cost ($)
Savic and Walters (1997) [1]	18	10	16	4	16	10	10	1	419,000
Cunha and Sousa (1999) [2]	18	10	16	4	16	10	10	1	419,000
Eusuff and Lansey (2003) [3]	18	10	16	4	16	10	10	1	419,000
Present work	18	10	16	4	16	10	10	1	419,000

Table 2 Evaluation number for different types of techniques used

Sl. No	Techniques used	Authors	Evaluation function numbers
1.	Genetic algorithm	Savic and Walters (1997) [1]	65,000
2.	Simulated annealing algorithm	Cunha and Sousa (1999) [2]	25,000
3.	Shuffled leap frog algorithm	Eusuff and Linsey (2003) [3]	11,155
4.	Shuffled complex algorithm	Liong and Atiquzzaman (2004) [8]	25,402
5.	Cuckoo Search algorithm	Naveen and Pankaj (2018)	15,300

5.2 Hanoi Water Distribution

The information related to different types of diameter available in the market, demand, and lengths for Hanoi network are taken from [7]. Totally 36 pipes with Hazen–Williams coefficient of 130 are used in networks. One reservoir with head 100 m and 32 demand nodes with minimum required pressure 30 m (Figs. 3, 4; Table 3).

6 Conclusions

The efficiency of Cuckoo Search algorithm was tested by solving two benchmark problems of water distribution networks, namely two-loop network and Hanoi water distribution network. The number of Function Evalutions obtained by cuckoo search algorithm when compared with Genetic Algorithm, Simulated Annealing algorithm showed significant improvement (i.e. found better solutions in few number of iterations). From the results, it is concluded that CSNET is efficient alternative evaluation model to optimize water distribution networks. CSNET can support the engineers to design of water distribution networks.

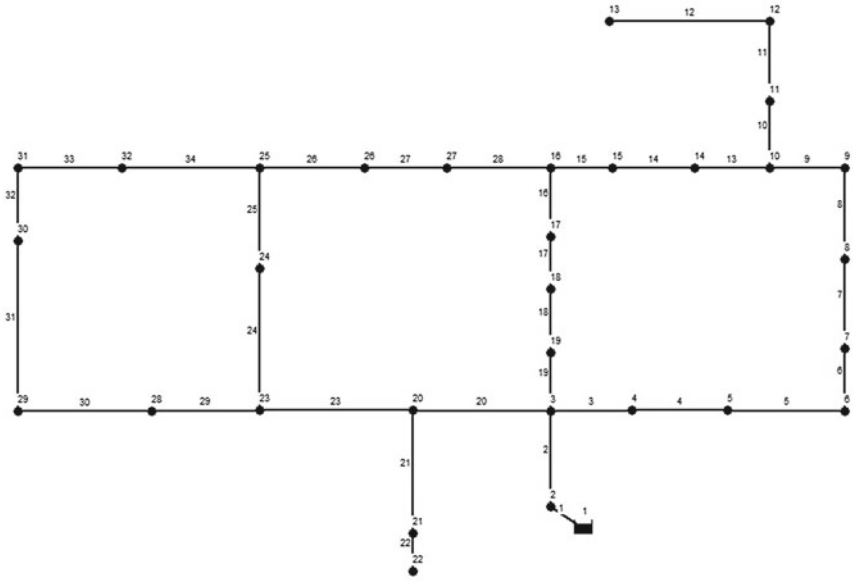

Fig. 3 Layout of Hanoi water distribution network [7]

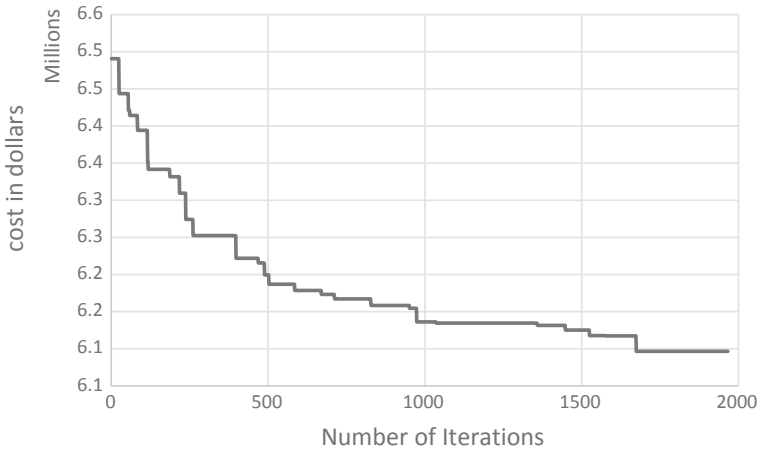

Fig. 4 Convergence graph for optimization of Hanoi water distribution network

Table 3 Optimal solutions for Hanoi water network for minimization of cost of diameters (inches)

Link	Savic and Walters (1997) [1]	Cunha and Sousa (1999) [2]	Liong and Atiquzzaman (2004) [8]	Current work Naveen and Pankaj (2018)
1	1016	1016	1016	1016
2	1016	1016	1016	1016
3	1016	1016	1016	1016
4	1016	1016	1016	1016
5	1016	1016	1016	1016
6	1016	1016	1016	1016
7	1016	1016	1016	1016
8	1016	1016	762	1016
9	1016	1016	762	1016
10	762	762	762	1016
11	609.6	609.6	762	762
12	609.6	609.6	609.6	609.6
13	508	508	406.4	609.6
14	406.4	406.4	304.8	508
15	304.8	304.8	304.8	304.8
16	304.8	304.8	609.6	304.8
17	406.4	406.4	762	304.8
18	508	508	762	406.4
19	508	508	762	609.6
20	1016	1016	1016	508
21	508	508	508	1016
22	304.8	304.8	304.8	508
23	1016	1016	762	304.8
24	762	762	762	1016
25	762	762	609.6	762
26	508	508	304.8	762
27	304.8	304.8	508	508
28	304.8	304.8	609.6	406.4
29	406.4	406.4	406.4	304.8
30	406.4	304.8	406.4	406.4
31	304.8	304.8	304.8	304.8
32	304.8	406.4	406.4	304.8
33	406.4	406.4	508	304.8
34	508	609.6	609.6	508
Cost ($)	6,073,000	6,056,000	6,220,000	6,096,565

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